Açai Palm Fruit (Euterpe oleracea Mart.) Pulp Improves Survival of Flies on a High Fat Diet

Reducing oxidative damage is thought to be an effective aging intervention. Açai, a fruit indigenous to the Amazon, is rich in phytochemicals that possesses high anti-oxidant activities, and has anti-inflammatory, anti-cancer and anti-cardiovascular disease properties. However, little is known about its potential anti-aging properties especially at the organismal level. Here we evaluated the effect of açai pulp on modulating lifespan in Drosophila melanogaster. We found that açai supplementation at 2% in the food increased the lifespan of female flies fed a high fat diet compared to the non-supplemented control. We measured transcript changes induced by açai for age-related genes. Although transcript levels of most genes tested were not altered, açai increased the transcript level of l(2)efl, a small heat-shock-related protein, and two detoxification genes, gstD1 and mtnA, while decreasing the transcript level of phosphoenolpyruvate carboxykinase (Pepck), a key gene involved in gluconeogenesis. Furthermore, açai increased the lifespan of oxidative stressed females caused by sod1 RNAi. This suggests that açai improves survival of flies fed a high fat diet through activation of stress response pathways and suppression of Pepck expression. Açai has the potential to antagonize the detrimental effect of fat in the diet and alleviate oxidative stress in aging.